

Structural Induction on Trees

Lecture Notes Transcription

Course: Interactive Theorem Proving (7CCSACTP)

1. Definitions & Recalled Concepts

Functions on Lists & Trees:

- `List-set :: 'a list  $\Rightarrow$  'a set`

`List-set [] =  $\emptyset$`

`List-set (x # xs) = insert x (List-set xs)`

- `Tree-set :: 'a tree  $\Rightarrow$  'a set`

`Tree-set Leaf =  $\emptyset$`

`Tree-set (Node l x r) = insert x (Tree-set l  $\cup$  Tree-set r)`

Inorder Traversal:

- `inorder :: 'a tree  $\Rightarrow$  'a list`

`inorder Leaf = []`

`inorder (Node l x r) = (inorder l) @ [x] @ (inorder r)`

Theorem to Prove:

`List-set(inorder t) = Tree-set t` (by induction on t)

2. Base Case ($t = \text{Leaf}$)

Goal: `List-set(inorder Leaf) = Tree-set Leaf`

Proof Steps:

1. `List-set(inorder Leaf) = List-set []` (by *subst inorder.simps(1)*)
2. `... =  $\emptyset$` (by *subst List-set.simps(1)*)
3. `... = Tree-set Leaf` (by *Tree-set.simps(1)*) $\checkmark$

3. Step Case ($t = \text{Node } t_1 \ a \ t_2$)

Induction Hypotheses:

- `I.H.1 : List-set(inorder  $t_1$ ) = Tree-set  $t_1$`
- `I.H.2 : List-set(inorder  $t_2$ ) = Tree-set  $t_2$`

Goal: `List-set(inorder(Node  $t_1 \ a \ t_2$ )) = Tree-set(Node  $t_1 \ a \ t_2$ )`

Auxiliary Lemma (Lem 1): `List-set(xs @ ys) = List-set xs  $\cup$  List-set ys`

Step-by-Step Equational Proof:

1. `List-set(inorder(Node  $t_1 \ a \ t_2$ )) = List-set(inorder  $t_1$  @ [a] @ inorder  $t_2$ )` (*inorder.simps(2)*)
2. `... = List-set(inorder  $t_1$ )  $\cup$  List-set([a] @ inorder  $t_2$ )` (by *Lem 1*)
3. `... = List-set(inorder  $t_1$ )  $\cup$  List-set[a]  $\cup$  List-set(inorder  $t_2$ )` (by *Lem 1*)
4. `... = Tree-set  $t_1$   $\cup$  List-set[a]  $\cup$  List-set(inorder  $t_2$ )` (by *I.H.₁*)
5. `... = Tree-set  $t_1$   $\cup$  insert a (List-set [])  $\cup$  List-set(inorder  $t_2$ )` (*List-set.simps(2)*)
6. `... = Tree-set  $t_1$   $\cup$  insert a  $\emptyset$   $\cup$  List-set(inorder  $t_2$ )` (*List-set.simps(1)*)
7. `... = Tree-set  $t_1$   $\cup$  {a}  $\cup$  Tree-set  $t_2$` (by *I.H.₂*)
8. `... = {a}  $\cup$  (Tree-set  $t_1$   $\cup$  Tree-set  $t_2$ )`
9. `... = Tree-set(Node  $t_1 \ a \ t_2$ )` $\checkmark$